P13 Level Complete

CRE132 worksheet

What you'll make

An end to the level. Collect coins and reach the big glowing stone at the end: the player stops, a sound plays, and a **Level Complete!** panel shows your score with **Next Level** and **Replay** buttons. **Next Level** fades into level 2 and keeps your score. At the end of level 2 the button says **Main Menu** and takes you to S06's menu.

You'll see that finishing a level is just another trigger, like coins and checkpoints, and why the next level's name belongs in the Inspector, not in the code.

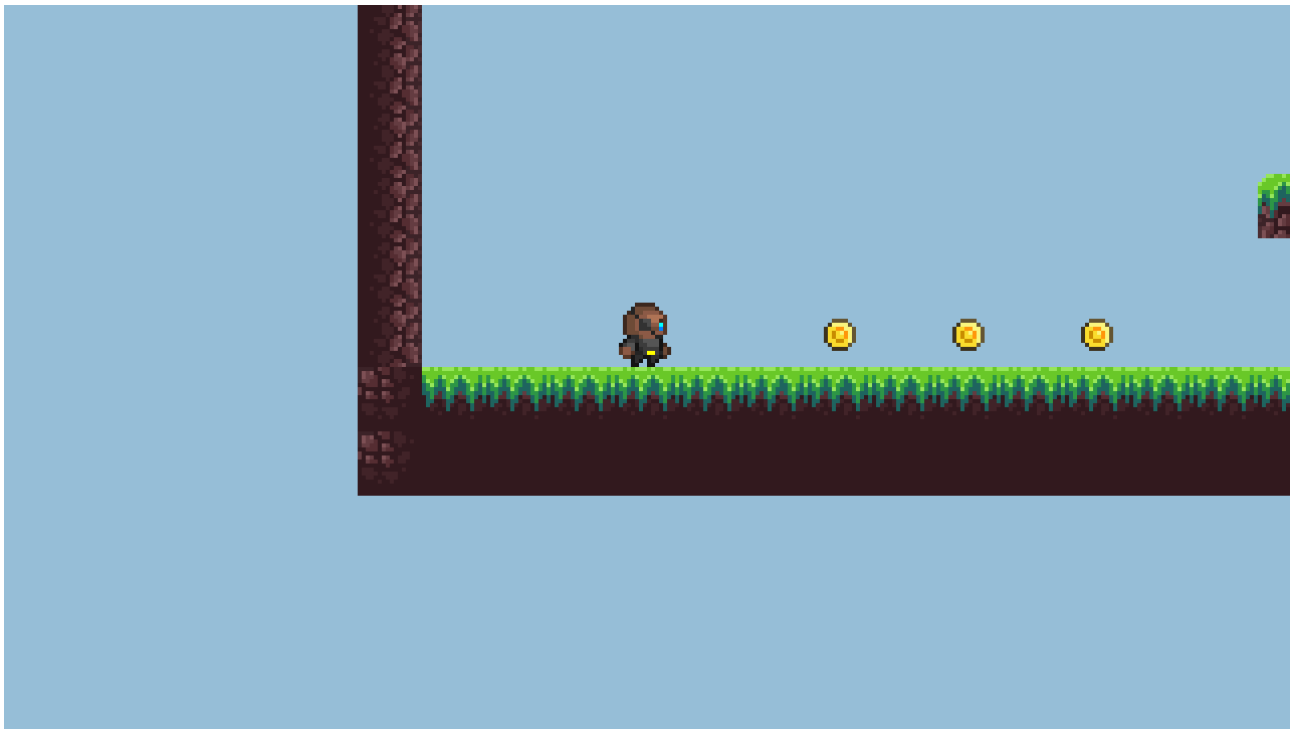


Figure 1: Level 1: the player at the start, with a row of coins ahead.

Before you start

1. Unzip `CRE132_P13_LevelComplete_Starter.zip`. In Unity Hub, choose **Add > Add project from disk** and pick the `CRE132_P13_LevelComplete_Starter` folder. Open it with Unity **6000.3.11f1**.
2. Open the scene `Assets/_game/Scenes/Platformer/P13_LevelComplete`.
3. Press **Play**. Run (A/D or arrows), jump (Space) and collect coins: the score goes up. Reach the glowing **Goal** stone at the end and nothing happens. Press Play again to stop.

The project has four scenes, all in the build list: level 1 (`P13_LevelComplete`), level 2 (`P13_LevelComplete_2`, in the same folder), and S06's menu scenes in `Assets/_game/Scenes/Shared`. Each level has the **GameManager**, **Scene Loader** and **AudioManager** prefabs, a **Goal** with a trigger collider, and a **Level Complete Canvas** whose **Level Complete Panel** is switched off. Inside the panel's **Box** are a **Title**, a **Score** text and two buttons.

Making a script: in the Project window, go to the folder, right-click and choose **Create > Scripting > MonoBehaviour Script**. Type the name straight away and press Enter. The name must match the class name in the code exactly, capital letters included.

1. The panel: LevelCompleteScreen

You make the panel's script first, because the goal script in part 2 talks to it. Create `LevelCompleteScreen` in `Assets/_game/Scripts/UI`:

```
// The Level Complete panel: shows the score and loads the next level (or the main menu
// after the last one).
using TMPro;
using UnityEngine;
using UnityEngine.EventSystems;
using UnityEngine.SceneManagement;

public class LevelCompleteScreen : MonoBehaviour
{
    [SerializeField] private GameObject panel;
    [SerializeField] private TMP_Text scoreText;
    [SerializeField] private GameObject firstButton;

    // Typed in the Inspector for each level; it must be in the build list.
    [SerializeField] private string nextScene;

    private void Start()
    {
        panel.SetActive(false);
    }

    public void Show()
    {
        panel.SetActive(true);
        scoreText.text = "Score: " + GameManager.Instance.Score;
        EventSystem.current.SetSelectedGameObject(firstButton);
    }

    public void LoadNextLevel()
    {
        SceneLoader.Instance.LoadScene(nextScene);
    }

    public void ReplayLevel()
    {
        GameManager.Instance.ResetScore();
        SceneLoader.Instance.LoadScene(SceneManager.GetActiveScene().name);
    }
}
```

What's going on:

- **`Start` hides the panel**, so it's never left showing by mistake. **`Show`** switches it on and fills in the score from the GameManager.
- **`SetSelectedGameObject(firstButton)`** highlights the first button, so a gamepad or the keyboard can press it straight away (S06).
- **`nextScene` is a string set in the Inspector.** Level 1's panel says level 2, and level 2's says the main menu. Adding a level 3 needs a new scene and a new name, not new code.

- **`LoadNextLevel`** uses S04's Scene Loader, so the level fades out. The GameManager survives the load (S02), so the score keeps adding up. **`ReplayLevel`** resets the score first, then reloads this level.

Add it: select **Level Complete Canvas** and add **Level Complete Screen**. Fill the slots:

- **Panel**: drag in **Level Complete Panel**.
- **Score Text**: open **Level Complete Panel > Box > Score** and drag in its child **Text**.
- **First Button**: **Next Level Button**.
- **Next Scene**: type `P13_LevelComplete_2` exactly.

Now connect the buttons. Select **Next Level Button** and find **On Click ()** on its Button. It already has one row with an empty object slot (if not, click **+**). Drag **Level Complete Canvas** into the slot and choose **LevelCompleteScreen > LoadNextLevel ()**. Do the same for **Replay Button**, choosing **LevelCompleteScreen > ReplayLevel ()**.

2. The goal: LevelGoal

Make a folder `Assets/_game/Scripts/P13_LevelComplete` and create `LevelGoal` in it:

```
// The end of the level. Touching it stops the player and shows the Level Complete screen.
using UnityEngine;

public class LevelGoal : MonoBehaviour
{
    [SerializeField] private LevelCompleteScreen completeScreen;
    [SerializeField] private AudioClip completeSound;

    private bool isReached;

    private void OnTriggerEnter2D(Collider2D other)
    {
        if (isReached || !other.CompareTag("Player"))
        {
            return;
        }
        isReached = true;

        // Stop the player: switch the controller off and stop running (it still falls if
        // in the air).
        BetterJumpController controller = other.GetComponent<BetterJumpController>();
        if (controller != null)
        {
            controller.enabled = false;
        }
        Rigidbody2D body = other.GetComponent<Rigidbody2D>();
        body.linearVelocity = new Vector2(0f, body.linearVelocity.y);

        if (completeSound != null)
        {
            AudioManager.Instance.PlaySound(completeSound);
        }
        completeScreen.Show();
    }
}
```

- **`OnTriggerEnter2D`** runs when the player touches the goal, just like a coin. The `!other.CompareTag("Player")` check ignores anything else.
- **`isReached`** makes sure it only happens once, even if the player touches the goal twice.

- **Stop the player's control**, or they could keep walking around behind the panel. Only the sideways speed is set to 0, so a player who lands on the goal mid-jump still falls to the ground.
- **The sound** is played through S05's AudioManager, then `completeScreen.Show()` shows the panel.

Add it: select **Goal** and add **Level Goal**. Drag **Level Complete Canvas** into **Complete Screen**. For **Complete Sound**, click the small circle and pick **GS2_Treasure_Chest_Unlock_Open** (it's in `Assets/_game/Audio/SFX`).

Test it: press Play and run to the goal. The player stops, the sound plays and the panel shows your score. **Replay** starts level 1 again with a score of 0. **Next Level** fades into level 2, but level 2 isn't set up yet.

3. Level 2

Save level 1 (Ctrl+S) and open `P13_LevelComplete_2`. It's built the same way, but its panel has a **Main Menu Button** instead of **Next Level Button**. Repeat parts 1 and 2 here (the scripts are already written, so you only add and fill them):

- On **Level Complete Canvas**, add **Level Complete Screen**: **Panel** is **Level Complete Panel**, **Score Text** is the **Text** inside **Score**, **First Button** is **Main Menu Button**, and **Next Scene** is `S06_Menus`.
- **Main Menu Button > On Click ()**: **LevelCompleteScreen > LoadNextLevel ()**. **Replay Button > On Click ()**: **LevelCompleteScreen > ReplayLevel ()**.
- On **Goal**, add **Level Goal**, with **Complete Screen** and **Complete Sound** as in level 1.

Save, open level 1 again and play right through. The score from level 1 carries on into level 2, and **Main Menu** at the end loads S06's menu.

> **The build list:** a scene can only be loaded by name if it's in the build list. Check it in **File > Build Profiles**, under **Scene List**: all four scenes should be there and ticked.

Check your work

- ☐ Reaching the goal stops the player, plays a sound and shows the panel with the score.
- ☐ Touching the goal again does nothing more.
- ☐ **Next Level** fades into level 2, and the score keeps counting up from where it was.
- ☐ **Replay** restarts the level with a score of 0.
- ☐ At the end of level 2, **Main Menu** goes to the menu.
- ☐ Compare with `CRE132_P13_LevelComplete_Finished` if something behaves differently.

Key points

- **A level is "complete" when the player touches a trigger**: the same idea as coins and checkpoints.
- **Store the next scene's name on the level, not in the code**, so adding level 3 is just a new scene and a new name.
- **Stop the player's control at the end**, or they can keep walking around behind the panel.

If something goes wrong

- **"Scene 'P13_LevelComplete_2' couldn't be loaded because it has not been added to the build settings":** the name in **Next Scene** is misspelt, or the scene isn't in the **Scene List**. Names must match exactly, capital letters and underscores included.
- **Nothing happens at the goal: Level Goal** isn't on the **Goal**, or the Goal's Box Collider 2D isn't a trigger.
- **NullReferenceException in LevelGoal:** the **Complete Screen** slot is empty.

- **NullReferenceException in Show:** the **Score Text** slot is empty, or there's no **GameManager** in the scene. Start from level 1, which has one.
- **The buttons do nothing:** their **On Click ()** rows have no object, or **No Function** is selected.

Try it

1. Make the player walk off the screen with a coroutine before the panel appears.
2. Add a timer and show the time on the panel as well as the score.
3. Add a level 3: duplicate level 2, change level 2's **Next Scene** to the new scene's name, and add it to the **Scene List**.